

## Mapping Key Biological Entities for Synthetic Circuit Modeling

To design synthetic gene circuits targeting senescent cells, we first identify and classify key biological entities involved in the cellular senescence process. These entities serve as inputs, regulators, or outputs in the circuit model.

Key Entities and Their Roles:

- **p53** — Tumor suppressor protein and transcription factor.  
Role: Can serve as an input sensor for DNA damage or stress response.
- **p21 (CDKN1A)** — Cyclin-dependent kinase inhibitor.  
Role: Used as a reporter or output for senescence induction.
- **p16INK4a (CDKN2A)** — Cell cycle inhibitor and senescence marker.  
Role: Potential promoter target to detect senescent cells.
- **NF-κB** — Transcription factor regulating SASP genes.  
Role: Acts as a regulatory node for SASP control in synthetic circuits.
- **IL-6 and IL-8** — Cytokines that are part of the SASP (senescence-associated secretory phenotype).  
Role: Serve as output indicators or feedback signals.
- **Caspase-3 and Caspase-7** — Apoptotic effector proteins.  
Role: Function as circuit effectors for selective senescent cell clearance.
- **BCL-2 family** — Anti-apoptotic proteins.  
Role: Serve as targets for inhibition to trigger cell death in senescent cells.

These biological entities represent the core molecular components that could be engineered into a synthetic gene circuit. In such a circuit, upstream sensors (like p53 or p16 promoters) detect senescence signals, while downstream effectors (like caspases) execute cell death, enabling selective elimination of senescent cells through GenAI-driven circuit design.